

Week	Dates (M-F)	Module	Topic	Learning Objectives		Assignments
				Lecture	Lab/Coding	Task Name
1	12 Jan - 16 Jan	Introduction and Data Wrangling	Introduction to Machine Learning and AI in Clinical Bioinformatics	(1) Key concepts and terminology, (2) Applications in clinical bioinformatics, (3) Course front matter	(1) Introduction to GitHub, (2) Environment setup (3) Python review / ML example (time permitting)	
2	19 Jan - 23 Jan		Data Preprocessing and Feature Engineering	(1) Data cleaning, (2) Normalization and standardization, (3) Feature selection/engineering, (4) Dealing with imbalanced data	(1) Data loading and cleaning, (2) Feature engineering examples, (3) Handling missing data, (4) Visualizations/data exploration	Data Curation
3	26 Jan - 30 Jan	Algorithms I	Supervised Learning I - Regression	(1) Introduction to regression models: linear, logistic, polynomial; (2) Use cases, (3) Metrics for evaluation: Mean squared error, R-squared, AUC (for logistic regression), (4) Regularization	(1) Implement linear models for continuous outcomes, (2) Implement a logistic model for binary outcomes, (3) Apply regularization techniques to manage overfitting	
4	02 Feb - 06 Feb		Supervised Learning II - Classification	(1) Introduction to classification algorithms: decision trees, random forest, k-NN, SVM, (2) Use cases, (3) Metrics for evaluation: accuracy, precision, recall, F1 score, confusion matrix, AUROC	(1) Implement classification models for binary outcomes, (2) Model evaluation, (3) Hyperparameter tuning	Supervised Learning
5	09 Feb - 13 Feb	Algorithms II	Unsupervised Learning I - Clustering	(1) Introduction to clustering: k-Means, hierarchical clustering, (2) Use cases (3) Metrics for evaluation: silhouette score, dendrograms, (4) Challenges: interpretation and determining number of clusters	(1) Implement clustering algorithms on data, (2) Visualization of clusters and interpretation, (3) Hyperparameter tuning	
6	16 Feb - 21 Feb		Unsupervised Learning II - Dimensionality Reduction	NO CLASS (Family Day Holiday) Lecture (Recorded): (1) Understanding the need for dimensionality reduction, (2) Dimensionality reduction techniques: PCA, t-SNE, UMAP, (3) Use cases, (4) Challenges: interpretability	(1) Implement dimensionality reduction on data, (2) Compare and contrast techniques, (3) Visualizations	Unsupervised Learning
7	23 Feb - 27 Feb	Midterm Test and Capstone Project	Review and Evaluation of Modules I-III	Midterm Review Lecture	Midterm Test Format: 20 multiple choice questions Material Covered: Weeks 1-6	
					(1) Introduction to Capstone Project (2) Co-working/Q&A	
8	02 Mar - 06 Mar	Applications	Applications I - Survival Modeling	(1) Introduction to survival analysis: Kaplan-Meier estimator, Cox proportional hazards model, (2) Handling censoring, (3) Applications: predicting time-to-event outcomes, (4) Metrics for evaluation (Concordance Index, log-rank test)	(1) Building survival models in Python, (2) Kaplan-Meier curve analysis and interpretation	Survival Analysis
9	09 Mar - 13 Mar		Applications II - Radiomics	(1) Introduction to radiomics: quantitative features from medical imaging, (2) Use case: AI using imaging data, (3) Key techniques, (4) Challenges: high-dimensionality data, small datasets, overfitting, reproducibility	(1) Image loading and handling, different types of imaging metadata, (2) Radiomics feature extraction (using PyRadiomics), (3) Building survival and classification models to predict patient outcomes based on radiomic features.	
10	16 Mar - 20 Mar		Applications III - Pharmacogenomics	(1) Introduction to pharmacogenomics: genetics and drug response, (2) Use case: AI for drug response prediction, (3) Key techniques, (4) Challenges: small datasets, heterogeneity, explainability	(1) Building models using pharmacogenomic data to predict drug response, (2) Feature importance analysis	
11	23 Mar - 27 Mar		Applications IV - Multiomics	(1) Key concepts: early vs. late fusion, cooperative learning; (2) Key techniques: random forests, dimensionality reduction, graph neural networks, (3) Ethical considerations: bias in ML models, data privacy, reproducibility	(1) Load and preprocess a multiomics dataset, (2) Train a random forest model on multiomics data (3) Compare early vs. late fusion approaches in classification tasks, (4) Implement a simple graph neural network for omics data integration	
12	30 Mar - 03 Apr	Capstone Project	Course Wrap-up and Capstone Project Presentations	(1) Course review: recap key concepts and discuss emerging trends/future directions in the field (~30min), (2) Student Feedback Questionnaire	Student presentations (~12X5min with room for technical blunder)	Capstone Project (Report + Presentation); in-class